Single neuron and population analysis of visual motion direction : effect of uncertaincy.

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Introduction

Studying the **internal representation** of natural features in the primary visual cortex (V1), like direction or spatial frequency, is crucial to understand how we perceive the external world. Research on 2D motion direction in **non-human primates** in particular when displaying **naturalistic stimuli** like MotionClouds reveals substantial diversity and multiple mechanisms within the neuronal population. The aim of this project is to examine how a large population of V1 neurons encodes stimulus direction and how this representation is modulated by the uncertainty using classical and decoding methods.

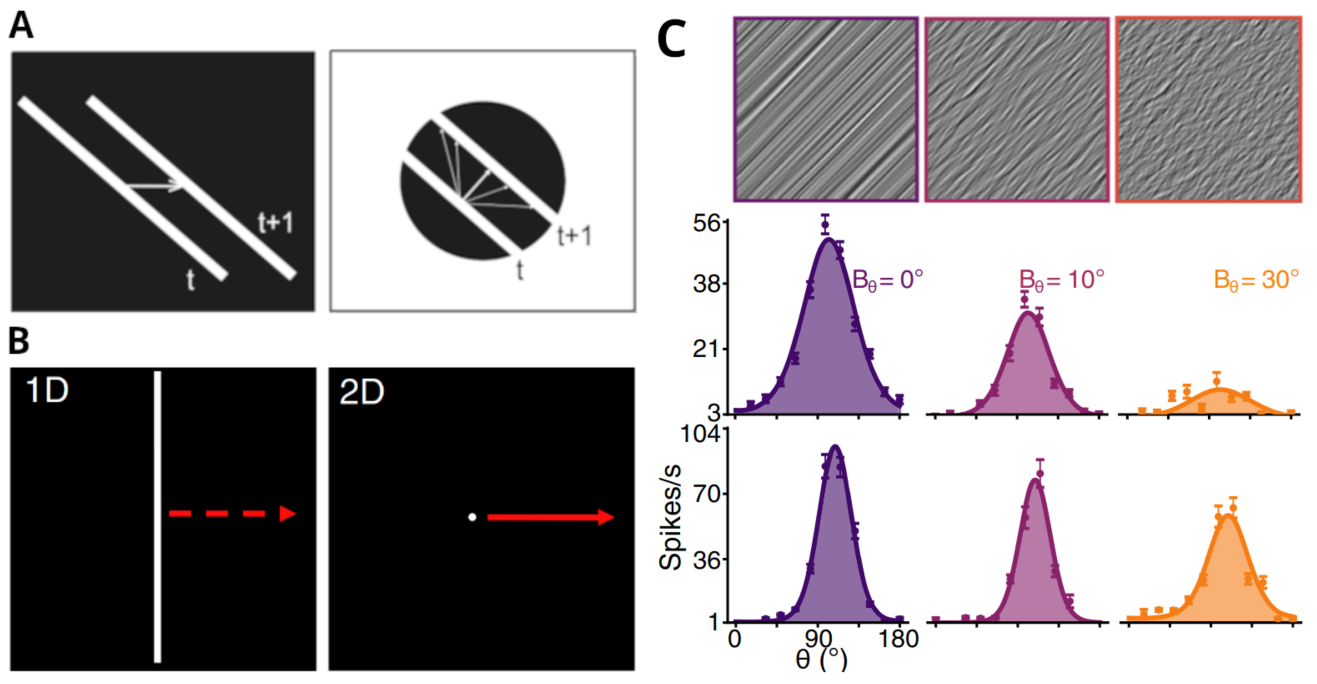


Figure 1: **A.** The aperture problem : information about motion is ambiguous when it is observed through a small aperture. **B.** Moving stimuli carrying mostly 2D (central spot, right) or 1D (very long line, left) information. **C.** From 1D to 2D, two subpopulation in cat area 17. (A and B adapted from A. Montagnini et al., Journal of Physiology, 2007; C adapted from H. Ladret et al., Nature Comm. Bio., 2023)

Methods

Neuronal recordings were made in V1 of one marmoset monkey using the Neuropixel technology during 2D motion presentation in twelve directions (θ) from $0°$ to $375°$, four levels of orientation bandwidth (B_θ; $0°$, $15°$, $30°$ and $45°$) and 4 levels of spatial frequency bandwidth (B_{sf}; 0.01, 0.175, 0.7 and 2.8 cpd) from the MotionClouds python library.

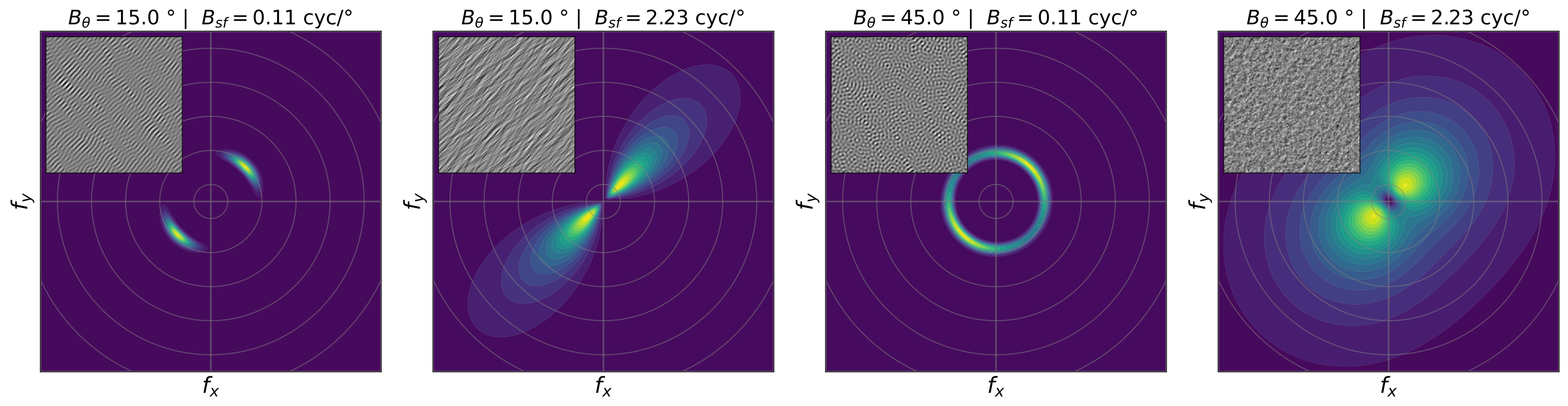


Figure 2: Fourier space representation of MotionClouds stimuli. The image is generated by the inverse fourier transform of the envelop after the application of a random phase.

Fitting in the Fourier space is based on a 2D functionnal space (f_r; f_θ) describe as follow :

$$f_r = e^{-\frac{1}{2}\times\log\frac{sf-sf_0}{B_{sf}}} \;;\; f_\theta = e^{\frac{\cos 2(\theta-\theta_0)}{4B_\theta^2}}$$

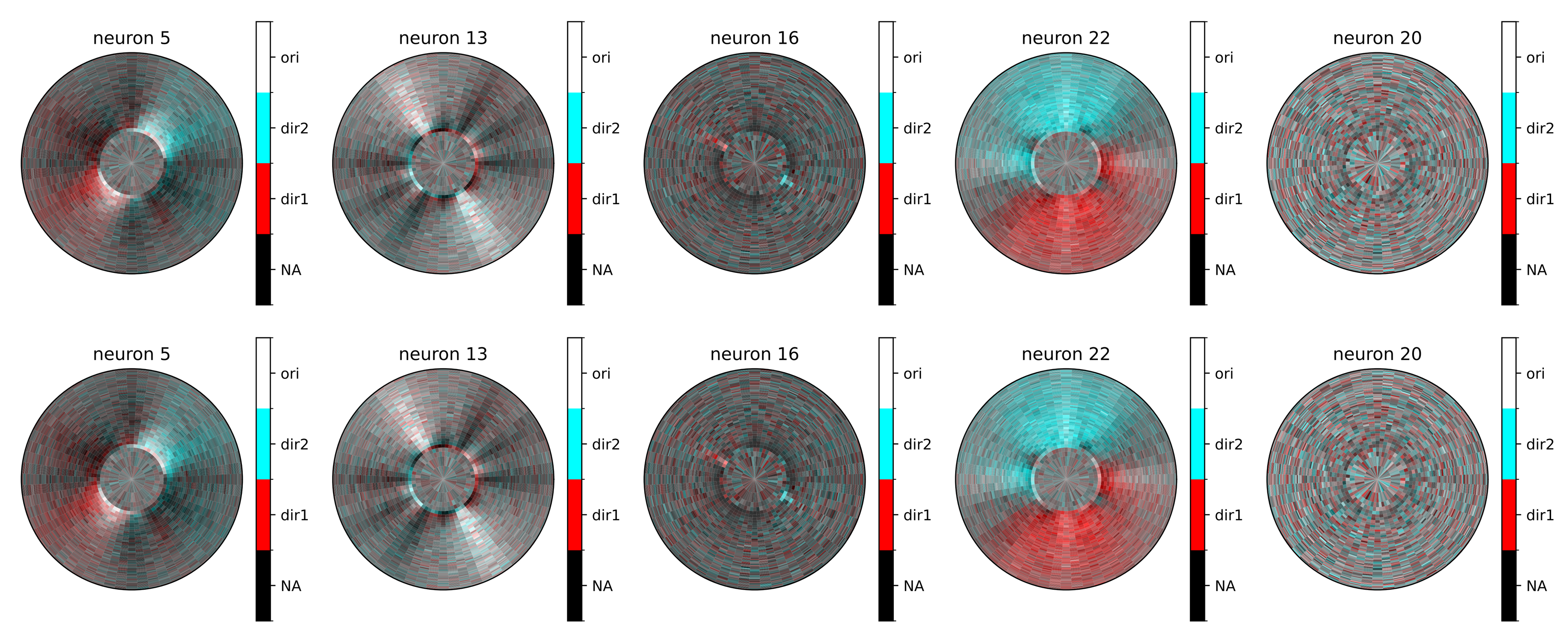
From this functionnal space, the tuning curve model is defined by the following equations parameters (κ_ϕ, κ_θ, θ_0...) in order to predicted the activity (A) of each neuron in each condition.

$$A_\theta = \kappa_\theta \times (\cos(2\times(\theta-\theta_0))-1)$$
$$A_\phi = \kappa_\phi \times (\cos(\theta-\theta_0)-1)$$
$$A = \exp\left(A_\theta + A_\phi + \log R_{amp}\right) + R_0$$

Based on our extracted parameters, orientation and direction selective indexs (OSI and DSI) are computed.
Decoding method based on Jazayeri and Movshon (2006). The likelihood function of the stimulus direction is equivalent to the sum of the likelihood functions of each neuron, which is equivalent to summing the agreement functions of each neuron weighted by the activity of each neuron during stimulus presentation. If we take W_i as the tuning curve of neuron i and A_i^s as the activity induced by simulus s. Then, L_i^s and L^s are respectively the likelihood function for a particular trial of the neuron i and the population.

$$\log L^s(\theta) = \sum_{i=0}^{N}\log L_i^s(\theta) = \sum_{i=0}^{N} A_i^s(\theta)\log W_i(\theta)$$

Results : single neuron analysis



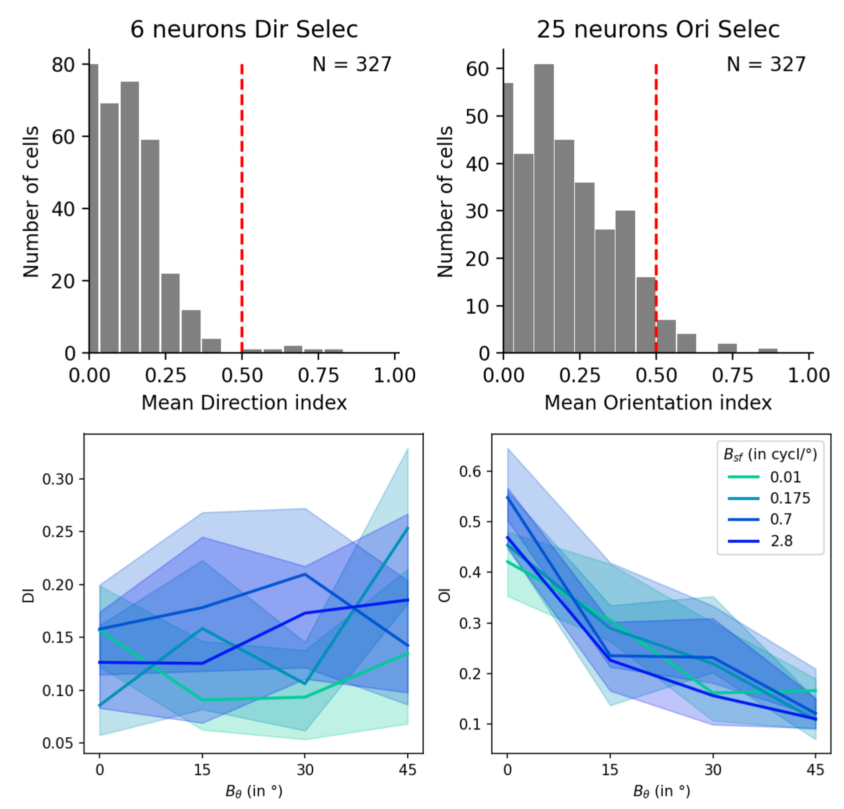


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Figure 3: Mean DSI and OSI in population.

Results : population decoding

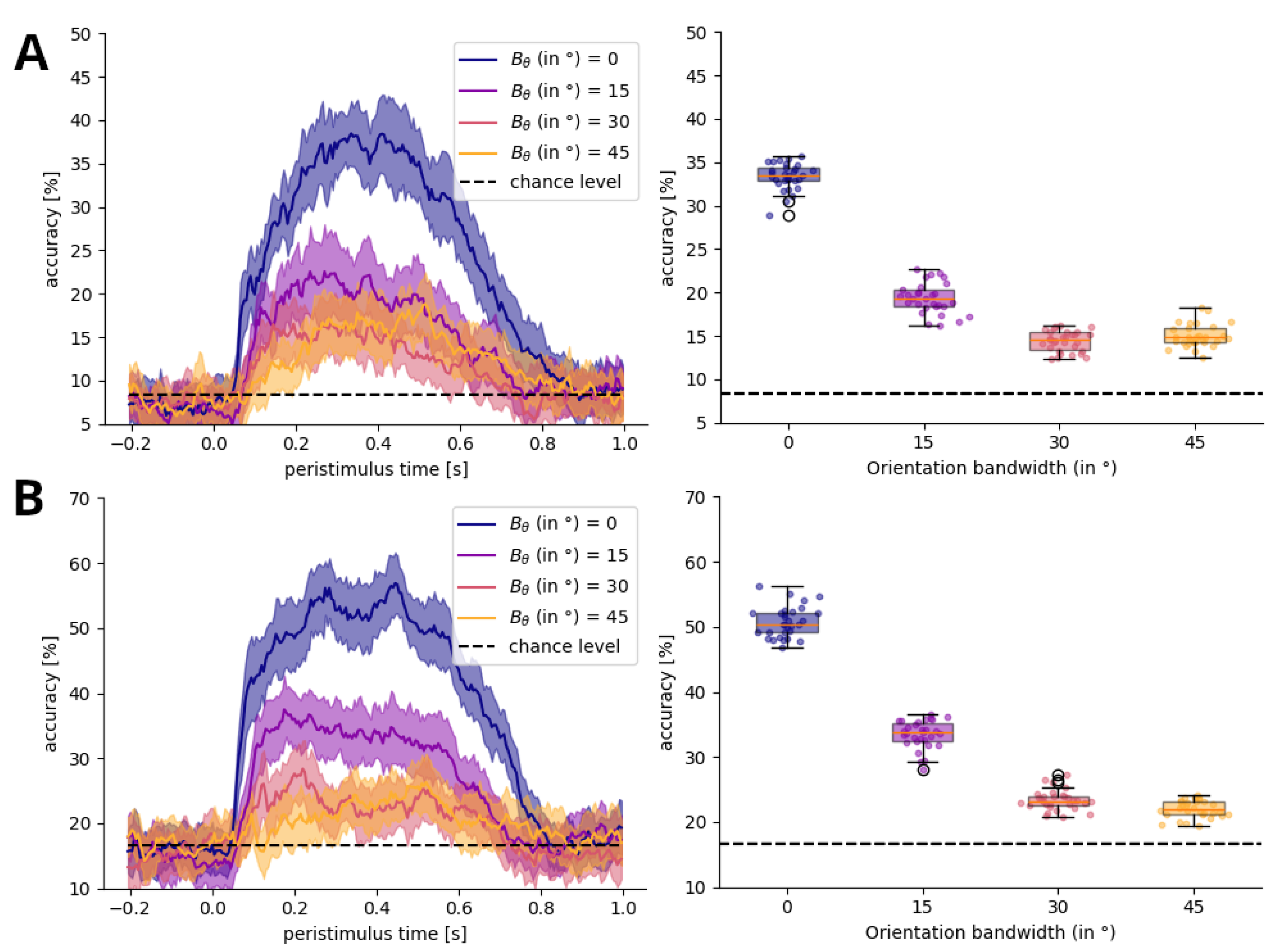

Figure 4: Decoding results for multiple level of orientation bandwidth. **A.** Decoding the direction of the stimulus. **B.** Decoding the orientaiton of the stimulus.

Discussion

This decoding approach clarifies the **representation of directional information** in the marmoset primary visual cortex, its stability accross time and its modulation by the precision level.